

BUSINESS PROBLEM STATEMENT

A leading retail organization aims to gain deeper insights into customer purchasing behavior to strengthen sales performance, enhance customer experience, and improve retention strategies. Recent shifts in consumer buying patterns across demographic groups, product categories, and purchasing channels (online and in-store) have highlighted the need for a more data-driven understanding of customer preferences and engagement.

Management seeks to identify the key factors influencing purchasing decisions and customer loyalty, including pricing strategies, promotional discounts, customer reviews, seasonal trends, and preferred payment methods. By leveraging consumer shopping data, the company intends to uncover actionable insights that can support targeted marketing initiatives, optimize product offerings, and improve overall business performance.

The primary business objective for this project is to answer the following question:

“How can consumer shopping data be utilized to identify behavioral trends, enhance customer engagement, and support more effective marketing and product decision-making?”

Project Deliverables

1. Data Preparation and Modeling (Python)

Clean, preprocess, and transform the raw dataset into a structured format suitable for analysis and reporting.

2. Data Analysis (SQL)

Design and organize the data into relational structures, simulate business transactions, and execute SQL queries to generate insights related to customer segmentation, purchasing behavior, loyalty patterns, and key sales drivers.

3. Data Visualization and Insights (Power BI)

Develop an interactive dashboard that presents key metrics, consumer trends, and performance indicators to support data-driven decision-making by stakeholders.

4. Business Report and Presentation

Prepare a professional report summarizing analytical findings, insights, and strategic recommendations. Create a presentation that clearly communicates results and actionable business opportunities to management and stakeholders.

5. GitHub Repository

Maintain a well-organized repository containing all project components, including Python scripts, SQL queries, datasets, documentation, and Power BI dashboard files.